

Table 7: Percentage Change in Language Likelihood Following Release, measured by wiki data from 2020 to 2024.3. Dash ("-") symbols in the table indicate missing data points due to the later release dates of some models, which means there is no available data for those specific time intervals.

Model	Base BPC	BPC Change at 3 months (%)	BPC Change at 6 months (%)	BPC Change at 9 months (%)	BPC Change at 12 months (%)	mean
OPT-13B	0.075	-0.623	0.105	-0.413	-0.818	-0.437
OPT-2.7B	0.082	-0.658	0.024	-0.528	-0.915	-0.519
LLaMA-7B	0.052	0.214	-0.425	0.379	-0.067	0.025
Pythia-12B	0.071	-0.578	-0.933	-0.859	-0.903	-0.818
Falcon-rw-1B	0.083	-1.272	-1.113	-1.131	-1.207	-1.181
Baichuan-13B-Chat	0.058	0.019	0.124	-0.125	-	0.006
LLaMA-2-13B	0.041	1.700	1.446	1.796	-	1.647
LLaMA-2-7B	0.050	0.774	0.349	0.517	-	0.547
LLaMA-2-7B-Chat	0.060	0.132	-0.186	-0.093	-	-0.049
Zhongjing-Base	0.050	0.756	0.269	0.569	-	0.531
InternLM-Chat-7B	0.080	-0.482	-0.301	-0.250	-	-0.344
Baichuan2-7B-Base	0.059	0.523	0.157	0.185	-	0.289
Baichuan2-7B-Chat	0.068	0.343	-0.035	-0.073	-	0.078
Mistral-7B-v0.1	0.053	0.381	-0.028	0.018	-	0.124
Phi-1.5	0.098	0.066	-0.125	-0.248	-	-0.102
Baichuan2-13B-Base	0.053	1.019	0.797	0.900	-	0.905
Baichuan2-13B-Chat	0.062	0.299	0.628	0.512	-	0.480
Colossal-LLaMA-2-7B-Base	0.065	-0.042	-0.199	-	-	-0.121
Qwen-14B-Chat	0.049	1.301	1.939	-	-	1.620
Qwen-7B	0.061	0.428	0.430	-	-	0.429
Qwen-7B-Chat	0.065	0.302	0.226	-	-	0.264
Skywork-13B-Base	0.050	0.075	0.276	-	-	0.175
ChatGLM3-6B	0.105	-0.187	-0.245	-	-	-0.216
Zephyr-7B-beta	0.059	-0.049	-0.033	-	-	-0.041
Yi-6B	0.058	-0.071	0.022	-	-	-0.024
Yi-6B-Chat	0.060	-0.028	0.023	-	-	-0.003
Qwen-1.8B	0.080	-0.131	-0.251	-	-	-0.191
Qwen-1.8B-Chat	0.091	-0.331	-0.496	-	-	-0.414
RWKV-v5-Eagle-7B	0.065	0.149	0.220	-	-	0.184
TinyLLaMA-1.1B-Chat-v0.6	0.070	-0.332	-0.473	-	-	-0.403
Phi-2	0.072	-0.037	-	-	-	-0.037

formance might overlook crucial aspects of cognitive and interpretative abilities that are essential for more complex applications.

M Details of Model Accuracy Peaks

Tab. 14 captures the peak performance periods of various models on Freshbench. It lists specific models along with their respective dates when they achieved the best performance.

N Analysis of Best-Achieved Accuracy and Cutoff Time

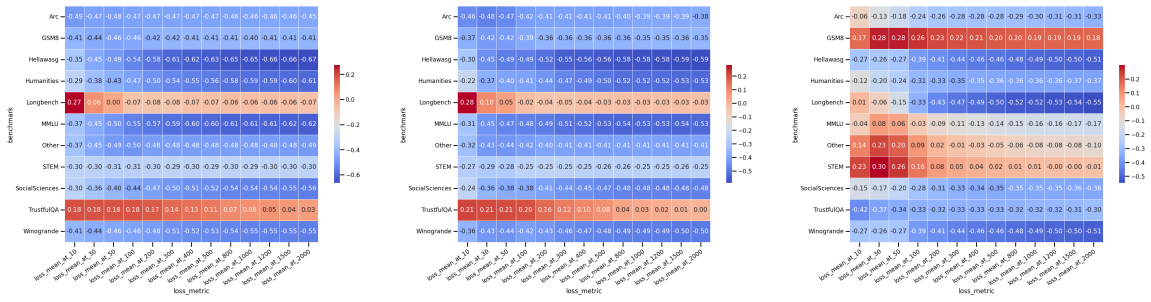
Tab. 15 shows an analysis covering the period from 2015 to 2024, with the year 2019 serving as a pivotal midpoint, a distinct pattern emerges from the performance data of various language models: all models reached their peak accuracy before 2020. This observation points to a notable trend toward "Nostalgia", where models demonstrate optimal

performance on data from the earlier part of the timeline analyzed. This tendency highlights the models' alignment with the informational characteristics prevalent prior to the latter half of the observed period, suggesting their training data may disproportionately reflect earlier times.

The peak accuracy periods analyzed are based on the average accuracy of all questions within three-month intervals, to provide a stable and robust assessment of each model's performance.

Table 9: Overview of text categories with their sources counts, and average lengths in one sample crawl.

Category	Source	Count	Length
Academic STEM	arXiv Mathematics	50	42850
	arXiv Computer Science	50	43336
	arXiv Physics	50	42580
	arXiv Statistics	50	42619
	arXiv Electrical Engineering and Systems Science	50	43027
Academic Non-STEM	arXiv Economics	50	42592
	arXiv Quantitative Finance	50	43969
	arXiv Quantitative Biology	50	42974
Code	GitHub	30	21849
Internet QA	Reddit	583	643
	Quora Health	20	1150
	Quora Movies	69	1349
	Quora Mathematics	70	1365
	Quora Technology	80	1674
Internet article	Wiki	220	4091
	Wattpad	38	6615
	BBC	45	4792
	Yahoo	41	5373



(a) STEM: base models

(b) non-STEM: base models

(c) Non-STEM: chat models

Figure 11: The heatmaps display the correlations by loss for a group of models across different text chunk lengths and different benchmark scores. Each row represents a specific truncation length, while the columns represent different benchmark score categories. The values in the cells indicate the corresponding correlation coefficients. (a) presents the correlations by loss for all 18 base models on STEM Academia texts. (b) presents the correlations by loss for all 18 base models on non-STEM Academia texts. (c) presents the correlations by loss for all 15 chat models on non-STEM Academia